

Abed Zahed

RF & Wireless Systems Engineer | Telecom & Wireless Sensing | Engineering Lead

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Nationality: Swedish | Location: Sharjah, UAE



EXPERIENCE

Consultant Engineer

Fun Robotics

11/2024 – 08/2025 📍 Dubai, UAE

- Coordinated between engineering teams, UAE TDRA, and training partners to ensure delivery alignment with UAE telecom regulatory requirements.
- Collaborated with TDRA to design and deliver technical workshops, including curriculum development and coordination with Etisalat Academy.
- Led a team to develop and deliver technical training on wireless communications and RF systems.

Tech Consultant

Mpya Sci & Tech

10/2020 – 08/2024 📍 Gothenburg, Sweden

- Deployed as engineering consultant across Ericsson and Veoneer client sites via Mpya Sci & Tech's professional-services model.
- Contributed to internal Mpya initiatives; held After-Work Committee Coordinator and Consultant Buddy roles supporting team culture and new consultant onboarding.

Microwave DSP Config Developer (Consultant)

Ericsson

03/2022 – 12/2023 📍 Gothenburg, Sweden

- Conducted system testing via FPGA channel emulator (ChEmu): selective fading, channel fading, cross-polar discrimination, modulation switch-up/down sensitivity, and noise level analysis.
- Led project for a team of 4 engineers, structuring testing methodologies aligned with ETSI, ANSI, and IEC standards.
- Implemented MRC and novel MRCx combining on XPIC channel modes for Ericsson MiniLink; achieved 2.7 dB signal power gain.

SUMMARY

RF and wireless systems engineer with 5+ years at Ericsson (led MRC/MRCx implementation, 2.7 dB signal gain; MATLAB, Python, and FPGA test bench for SISO/XPIC/MIMO) and Veoneer (65% to 98% production turnaround via root cause analysis). M.Sc. Wireless, Photonics and Space Engineering, Chalmers; B.Sc. Electrical Engineering, AUS; 4 IEEE publications in RF sensing.

Process Engineer (Consultant)

Veoneer / Magna International

11/2020 – 01/2022 📌 Vårgårda, Sweden

- Improved machine performance from 65% to 98% by leading a team of 3 optics engineers in troubleshooting, root cause analysis, and documentation.
- Directed incident and problem resolution for camera production; coordinated cross-functional investigations with production, engineering, and maintenance teams.
- Tracked production efficiency and process capability KPIs via Power BI dashboards and Python pipelines; adopted org-wide for yield monitoring.

Space Systems Research (M.Sc. Program)

Onsala Space Observatory / Chalmers University

2018 – 2019 📌 Onsala, Sweden

- Completed research training at Onsala Space Observatory covering satellite communications, link budgets, and space systems engineering as part of M.Sc. in Wireless, Photonics and Space Engineering at Chalmers.
- Operated GNSS tracking and analysis systems, satellite positioning equipment, and antenna dish control systems in an active deep space observatory.
- Conducted deep space data processing and analysis using space-grade RF measurement and data acquisition equipment.

Master's Thesis: RF Researcher

RISE Research Institutes of Sweden

02/2019 – 06/2019 📌 Borås / Gothenburg, Sweden

- Increased detection sensitivity by 3x via patch antenna research for transformer partial discharge monitoring.
- Designed and tested a prototype patch antenna using VNA, spectrum analyzer, and signal generators; calibrated test setup demonstrated performance comparable to standardized measurement setups.
- Measured and characterized radiation patterns for the patch antenna prototype; validated gain, return loss, and 50Ω impedance matching across the UHF frequency range.

Robotics Engineer

Fun Robotics

06/2016 – 08/2017 📌 Dubai, UAE

- Taught wireless communication fundamentals, protocol configuration, embedded systems, and RF signal concepts to students age 7-20.
- Designed comprehensive teaching curricula for diverse robotics and software platforms; adopted as educational resources for the academic industry.
- Designed and manufactured robotic solutions as part of the product development team.

SKILLS

CORE COMPETENCIES

Technical Training & Knowledge Transfer

Regulatory Compliance (ETSI / ANSI / IEC)

Cross-functional Team Leadership

TDRA / UAE Telecom Regulatory Coordination

TELECOM & ENGINEERING

RF engineering

wireless systems

Wireless & Microwave Systems (LTE/5G, Radio Links)

RF Testing, Validation & Performance Analysis

RF Lab Equipment and Instrumentation (VNA, Spectrum Analyzer, Signal Generators, Network Analyzers)

Circuit and PCB Design (RF antenna PCBs, fractal antennas, diagnostic test setups; Altium)

RF Front-End Systems (LNAs, PAs, mixers, filters, T/R modules; systems-level design)

Digital Signal Processing (DSP)

Machine Learning (ANN, SVM, RF sensing classification)

MIMO / XPIC / SISO channel modeling

TOOLS & PLATFORMS

Odoo ERP (Inventory & POS)

Electronics Testing and Diagnostics (RF systems, channel emulation, performance validation)

FPGA channel emulation (ChEmu)

MATLAB

Python (test automation, data analysis)

Simulink

HFSS

ADS, CST Microwave Studio (EM simulation)

Visiting Researcher

American University of Sharjah

01/2015 – 10/2015 📌 Sharjah, UAE

- Conducted advanced research on partial discharge analysis using Hilbert fractal antennas; achieved 97% classification accuracy.
- Co-authored 4 IEEE journal and conference publications in RF sensing and diagnostic antenna design.
- Designed PCB-based RF antennas with 50Ω coaxial feeds; applied transmission line theory for impedance matching and optimized RF circuit feed networks.

EDUCATION

M.Sc. in Wireless, Photonics, and Space Engineering

Chalmers University of Technology

2017-2019 📌 Gothenburg, Sweden

B.Sc. in Electrical Engineering (Minor: Computer Engineering)

American University of Sharjah

2011-2015 📌 Sharjah, UAE

KEY ACHIEVEMENTS

◆ MRC/MRCx Implementation

Led implementation on Ericsson MiniLink products achieving 2.7 dB gain in received signal power, effectively doubling received power.

◆ UAE Telecom Training Delivery

Designed and delivered technical training with UAE TDRA in partnership with Etisalat Academy, supporting regulatory capability building.

◆ Production Performance Turnaround

Improved camera production line performance from 65% to 98% at Veoneer/Magna by leading a team of 3 optics engineers through root cause analysis and corrective action.

◆ RF Sensing Research

Increased detection sensitivity 3x via patch antenna research at RISE Sweden; co-authored 4 IEEE publications in RF sensing and diagnostic antenna design.

◆ Operational Analytics

Developed Power BI dashboards adopted org-wide at Veoneer for production yield monitoring and Python-automated data analysis pipelines.

LANGUAGES

Arabic	●●●●●	Native
English	●●●●●	Full Professional
Swedish	●●●●●	Elementary (improving)
Spanish	●●●●●	Elementary